Final Project

Machine Learning Usage in Breast Cancer Prediction

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Abstract

Breast cancer remains as one of the leading causes of cancer related mortality, and early, accurate diagnosis is critical for improving patient outcomes. Traditionally, patients with abnormal radiology results have to return for biopsies, which are invasive, costly, and frequently lead to unnecessary procedures for benign cases. In this study we investigate whether statistical and machine learning methods can effectively predict whether a breast mass is benign or malignant using radiological measurements; machine learning can potentially reduce unnecessary biopsies if it can make these predictions. We utilized the publicly available breast cancer dataset from Kaggle, comprising 569 patient observations with 32 features that include mean and standard error, and radiological measurements like radius, texture, perimeter, area, and smoothness. Diagnosis was the target variable that classifies tumors as benign or malignant.

After preprocessing and exploratory data analysis we applied several statistical classification techniques like logistic regression, random tree, and decision tree to model the relationship between radiologic features and diagnostic outcomes. Model performance was evaluated using accuracy, ROC curve (AUC), and sensitivity. Our results demonstrate properly tuned statistical models can achieve high predictive performance, indicating that radiology features contain significant information to distinguish benign and malignant tumors. These findings support the potential use of statistical prediction models as a noninvasive decision support tool in clinical settings, possibly minimizing unnecessary biopsies.

Keywords

Breast cancer; Regression; Classification; Machine Learning Models; Diagnosis

Introduction

Breast cancer represents one of the most pressing global health challenges, accounting for 2.3 million new diagnoses in 2020 and ranking as the most commonly diagnosed cancer among women worldwide (Ferlay et al., 2024). Despite advances in screening and treatment, its prevalence underscores the urgent need for continued research into both prevention and improved diagnostic methods. Yet the scale of the disease tells only part of the story; its impact extends far beyond incidence statistics.

Beyond the statistics, a cancer diagnosis profoundly affects individuals and their families. Patients often face not only the physical burden of treatment but also significant psychological, social, and financial consequences. The uncertainty surrounding prognosis, the disruption of daily life, and the emotional toll of navigating complex medical decisions highlight the importance of accurate and timely diagnosis. Many patients struggle with anxiety, depression, social isolation, and productivity loss (Cayrol et al., 2019). As Foster et al. (2022) note in *Understanding the Impact of Cancer Diagnosis and Treatment on Everyday Life*, women diagnosed with cancer typically experience a reduction in self-reported quality of life, which may take an average of 12 months to return to baseline. This trajectory underscores that a cancer diagnosis is not a single event but a long and complex journey for survivors. Given these emotional and practical challenges, the timing and accuracy of diagnosis become especially critical.

Clinical outcomes are closely tied to the stage at which breast cancer is detected. Early-stage detection is associated with cure rates as high as 90%, whereas diagnosis in intermediate stages reduces cure rates to 50–80%. Late-stage detection makes curative treatment exceedingly difficult (Wen et al., 2024). The first line screening tool, the clinical breast exam (CBE), involves manual examination by a trained clinician. However, its effectiveness remains debated. A recent meta-review found no direct evidence that CBEs reduce breast cancer related mortality, though it did reveal important insights into CBE performance. Overall, CBEs demonstrated specificity comparable to mammography but generally lower sensitivity. Notably, among women of Asian descent and younger women, CBE sensitivity approached that of mammography, suggesting that performance varies across demographic groups (Ngan et al., 2020). Other first-line screening methods for early breast cancer detection, such as ultrasound and mammography have demonstrated pitfalls as well. Radiologists who read and analyze the medical imaging from these screening methods are inherently subjective and human errors can occur - a survey found that in an eight-hour workday radiologists commonly suffer from eye fatigue and short analysis time, leading to unstable findings in the imaging (Wen et al., 2024). Together, these limitations highlight the need for complementary diagnostic approaches that can support clinicians and reduce variability in early detection.

Introduced in 2001, computer-aided detection (CAD) systems for mammography have been used extensively by American radiologists. Traditional CAD systems relied on rule-based algorithms and as a result produced high false-positive rates and radiologist recalls. More recently, artificial intelligence (AI) - CAD systems have emerged as a promising alternative. They utilize machine learning and deep learning techniques to analyze imaging data with greater precision. Unlike traditional CAD systems, AI-CAD systems can learn from complex datasets and continue to

improve into the future. AI-CAD systems are utilized to reduce risk of human error and overcome limitations of traditional diagnostic methods (Le et al., 2019). Machine learning techniques in AI-CAD systems today are used to support current systems to enhance survival rates by aiding the decision-making process (Yassin et al., 2018). As these technologies continue to evolve, they offer a valuable complement to traditional screening methods, reinforcing the need for robust, interpretable statistical models that can further strengthen diagnostic accuracy and support clinicians in early cancer detection.

In this study, we apply machine learning models including regression and tree based methods to a dataset of breast tumor characteristics to predict whether tumors are benign or malignant. By examining statistical relationships among tumor features, these machine learning modeling methods provide a transparent and interpretable framework to support clinical decision-making and enhance diagnostic accuracy. This approach demonstrates how statistical modeling can complement traditional diagnostic practices, offering a pathway toward more precise and patient-centered cancer care.

Literature Review

Research on breast cancer consistently highlights the importance of accurate tumor characterization for guiding clinical decision-making and improving patient outcomes. In a study published in the *New England Journal of Medicine*, Hartmann et al. (2006) investigated long-term outcomes in women with early-stage breast cancer, focusing on how different treatment regimens influenced recurrence and survival. The study followed a large cohort of patients who received various combinations of surgery, radiation, and therapies over multiple years. The authors found that treatment choice significantly affected patient prognosis, with some combinations reducing recurrence rates and improving overall survival. Importantly, the study emphasized that individual tumor characteristics—such as size, histologic type, and hormone receptor status—were key factors in determining the most effective treatment strategies. These findings underscore the importance of accurately classifying breast tumors as benign or malignant. Accurate classification is not only clinically relevant but also foundational for developing predictive models or statistical algorithms aimed at prognosis, recurrence risk, or therapy response. By demonstrating the impact of tumor characteristics and treatment decisions on patient outcomes, Hartmann et al.'s study provides a strong clinical rationale for breast cancer classification research, including regression-based and machine learning approaches, as a tool to support evidence-based, personalized care.

Further evidence of breast cancer's complexity comes from epidemiological research examining risk-factor heterogeneity across tumor subtypes. In a large population-based study published in Cancer Epidemiology, Biomarkers & Prevention, Li et al. examined how well-known breast-cancer risk factors (such as age, reproductive history, family history, hormone use, and other factors) are associated with the risk of developing different types of invasive breast cancer. By analyzing a large population, they stratified breast cancers not just as a single disease but by subtype, then assessed how each risk factor correlates with subtype-specific incidence. Their results showed that many traditional risk factors did not affect all subtypes equally. For example, some reproductive or hormonal risk factors were more strongly linked to certain subtypes than others. This suggests that invasive breast cancer is heterogeneous: different types may have different risk-factor profiles. The study underscores the importance of distinguishing between breast cancer subtypes, rather than treating all breast cancers as a single entity, when studying risk factors, prognosis, or when building predictive models.

Collectively, these findings illustrate that breast cancer is not only clinically diverse but also diagnostically challenging. This has prompted a shift toward computational methods capable of supporting clinicians in making more accurate and timely assessments. In recent years, machine-learning approaches have gained significant attention for their potential to improve breast cancer diagnosis, and a growing body of literature now evaluates their performance across imaging and clinical datasets.

A study by Hassan et al. (2023) compared machine learning techniques combined with Least Absolute Shrinkage and Selection Operator (LASSO) for breast cancer detection. The study specifically compared Random Forest (RF), K-Nearest Neighbors (KNN) and Multilayer

Perception (MLP) with all features and with features selected by LASSO. The results of the study show that Random Forest performed the best in accuracy scoring at 90.86%.

Similarly, Duan et al. (2024) developed and tested multiple models for predicting tumor diagnosis and prediction of metastasis development. Methods such as Support Vector Machine, Extreme Gradient Boosting, Gradient Boosting, Decision Tree, and Logistic Regression and Random Forest were used. In this study the Random Forest method demonstrated the greatest performance among these models with an accuracy of 93.6%, an F1-score of 88.9% and an AUC value of 91.3%.

Additionally, a research study by Faisal et al. (2018) compared multiple machine learning techniques including MLP, Decision Tree, SVM, Neural Network, Gradient Boosted Tree, and Naive Bayes on their ability to model the classification of diagnosing early stage breast cancer. In this study the Gradient - Boosted Tree achieved an accuracy rate of 90%, outperforming all other methods.

Taken together, the existing literature demonstrates both the clinical importance and the analytical complexity of breast cancer diagnosis. The reviewed studies highlight substantial variation in tumor characteristics and risk-factor profiles, underscoring the need for diagnostic tools that can accommodate this variability. Machine-learning research further shows that computational models, particularly ensemble-based and tree-based methods, consistently achieve strong performance in classifying breast tumors and predicting disease progression. At the same time, many of these high-performing models function as black-box systems, offering limited interpretability for clinicians who must understand and trust the basis of diagnostic decisions. These considerations point to the value of evaluating both interpretable regression-based models and more flexible tree-based approaches. With this in mind, this study aims to contribute to ongoing efforts to develop diagnostic tools that are both accurate and clinically meaningful.

Methodology

This study utilized the Breast Cancer Diagnostic data set. The goal was to apply regression models to classify tumors as benign or malignant based on their physical characteristics. The dataset consisted of identifiers, diagnosis labels, and 569 observations with 30 numeric features computed from digitized images of a fine needle aspirate (FNA) of a breast mass. These features describe characteristics of the cell nuclei, such as radius, texture, perimeter, and area. For this analysis, the dataset was subset to focus on the "mean" value features, excluding standard error and "worst" metrics to reduce dimensionality and avoid multicollinearity.

To address potential non-linearity, we assessed the distribution of all continuous predictors. Variables with high skewness (>1) were targeted for logarithmic transformation $y=\log(x+1)$ to normalize their distributions. Given the different units of measure across features, all numeric predictors were standardized using Z-score scaling (mean = 0, standard deviation = 1). The data was split into 70% training and 30% testing sets using random sampling to allow for unbiased model validation.

In the modeling strategy, we employed a multi-stage modeling approach, moving from interpretable linear models to complex machine learning algorithms.

1. Logistic Regression established a baseline for classification. This method was able to quantify the relationship between individual features and the probability of malignancy.
2. Decision Tree was built to capture potential non-linear relationships that logistic regression might miss.
3. Random Forest algorithm aggregated predictions from 500 decision trees. We applied a Random Forest algorithm to maximize predictive accuracy.

Experimentation and Results

The exploratory analysis of the Breast Cancer Wisconsin dataset revealed the characteristic critical for modeling. The initial examination of the target variable diagnosis revealed 357 benign cases and 212 malignant cases (Figure 1). While variables such as `smoothness_mean` and `symmetry_mean` followed a normal distribution, geometric features like `area_mean` and `compactness_mean` exhibited significant right-skewness (Figure 2). After log-transformation, `compactness_mean_log` and `concavity_mean_log` approximated normality (Figure 3). The correlation matrix showed substantial multicollinearity among the size-related features, `radius_mean`. `Perimeter_mean` and `area_mean` displayed correlation coefficients nearing 1.0 (Figure 4).

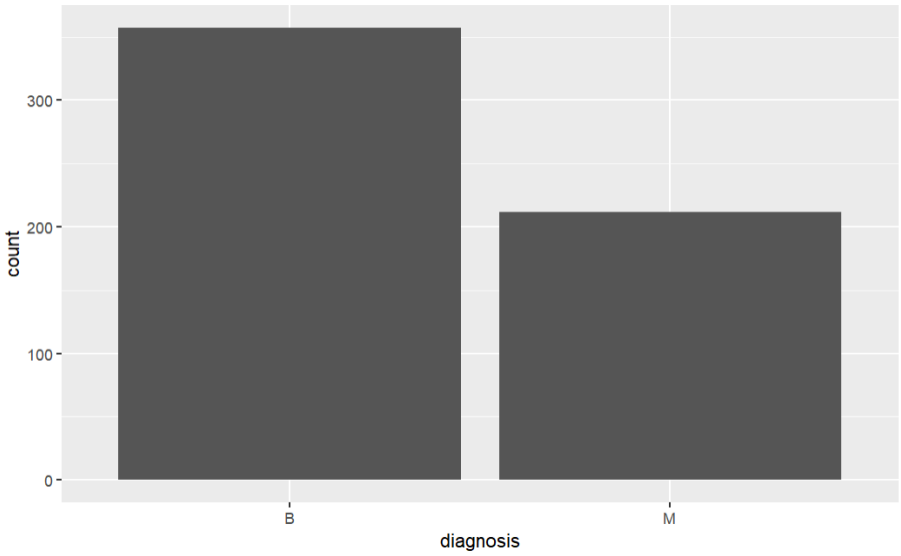


Figure 1: Distribution of the target variable

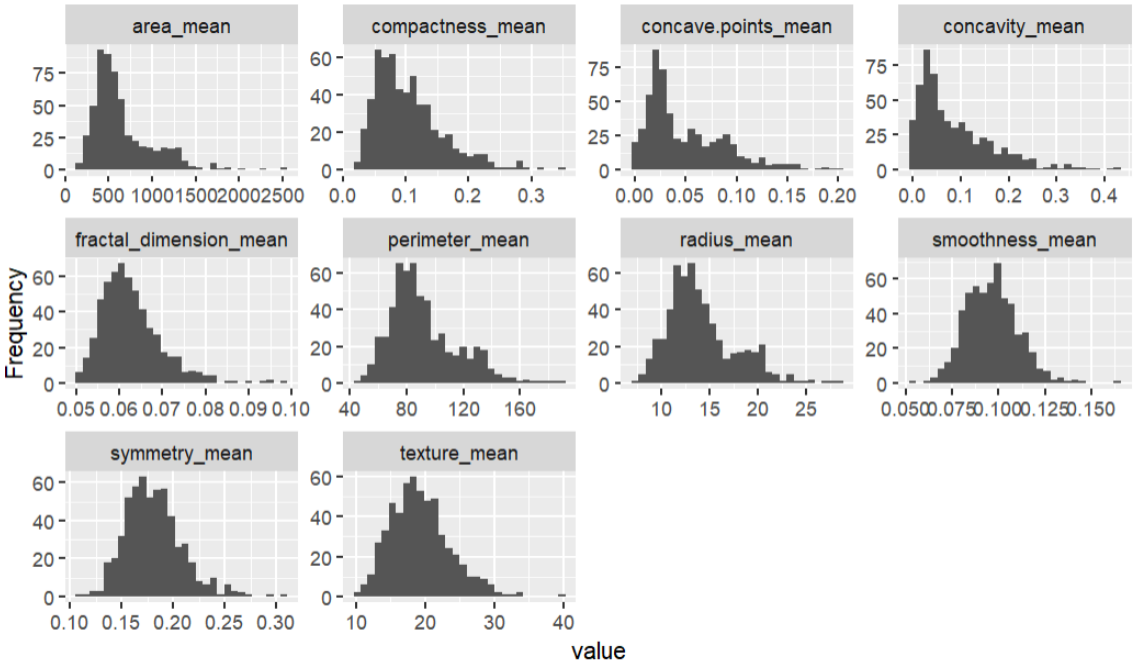


Figure 2: Distribution of all other variables

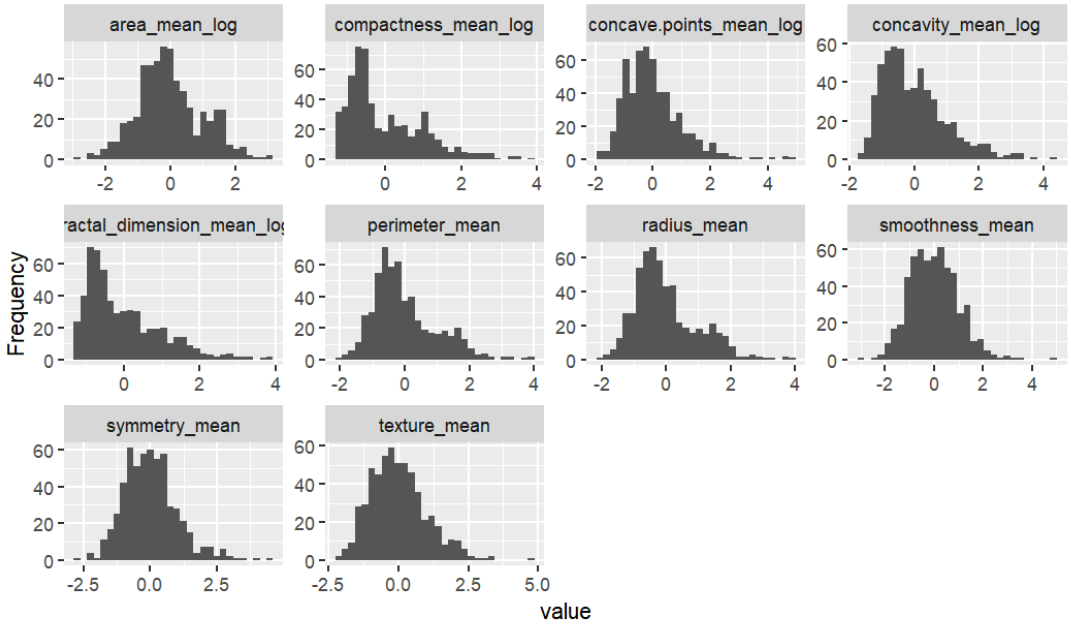


Figure 3: Distribution of all other variables after log transformation

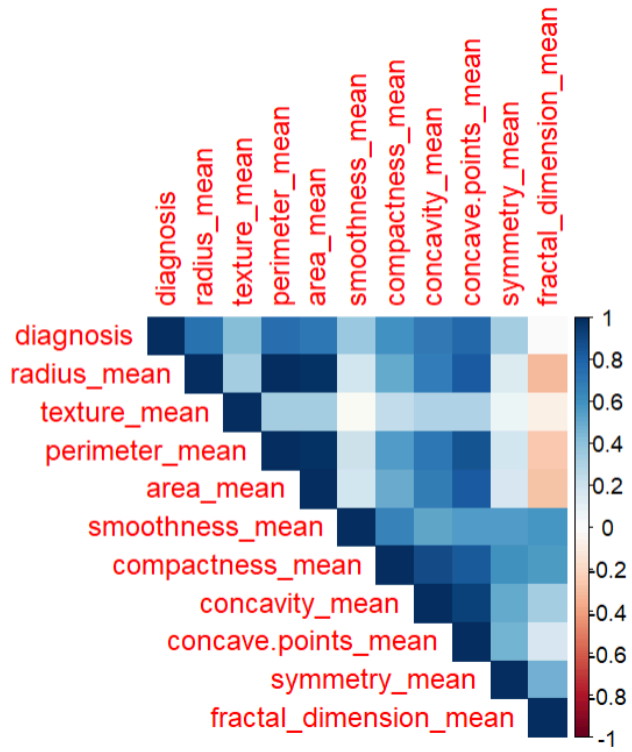


Figure 4: Correlation Matrix

Two logistic regression models were developed to predict the probability of malignancy. Model 1 incorporated all predictor variables, showed a strong statistical fit by reducing the Null Deviance from 51.85 to a Residual Deviance of 108.49. In this context, 'texture_mean' showed as the most statistically significant predictor with $z = 4.876$, and $p < 0.001$ (Table 1). To improve

interpretability, Model 2 only used `texture_mean`, `smoothness_mean` and `compactness_mean_log`. This model resulted in a high Residual Deviance of 142.6. Both `texture_mean` ($p < 0.001$) and `compactness_mean_log` ($p < 0.001$) are highly significant, while `smoothness_mean` has loose statistical significance of $p = 0.132$ (Table 2).

```
Call:
glm(formula = diagnosis ~ ., family = binomial, data = train)

Coefficients:
                            Estimate Std. Error z value Pr(>|z|)
(Intercept)                 -1.5541     0.6641  -2.340   0.0193 *
radius_mean                 14.1131    13.6360   1.035   0.3007
texture_mean                 1.5852     0.3251   4.876 1.08e-06 ***
perimeter_mean             -18.3830    16.4158  -1.120   0.2628
area_mean_log                6.3504     6.0730   1.046   0.2957
smoothness_mean              1.0525     0.5954   1.768   0.0771 .
concavity_mean_log           0.1463     1.3700   0.107   0.9150
fractal_dimension_mean_log   1.2710     0.8356   1.521   0.1282
compactness_mean_log         2.4628     1.3549   1.818   0.0691 .
symmetry_mean                0.3868     0.3519   1.099   0.2717
concave.points_mean_log     -0.1127     0.7057  -0.160   0.8732
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 515.85  on 398  degrees of freedom
Residual deviance: 108.49  on 388  degrees of freedom
AIC: 130.49

Number of Fisher Scoring iterations: 8
```

Table 1: Model 1 result

```
Call:
glm(formula = diagnosis ~ texture_mean + smoothness_mean + compactness_mean_log,
    family = binomial, data = train)

Coefficients:
                    Estimate Std. Error z value Pr(>|z|)
(Intercept)          -0.9738     0.2379  -4.092 4.27e-05 ***
texture_mean          1.3968     0.2749   5.081 3.76e-07 ***
smoothness_mean      -0.4076     0.2707  -1.506    0.132
compactness_mean_log  4.7741     0.5636   8.471  < 2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 515.85  on 398  degrees of freedom
Residual deviance: 142.65  on 395  degrees of freedom
AIC: 150.65

Number of Fisher Scoring iterations: 7
```

Table 2: Model 2 result

A Decision Tree was applied to capture non-linear relationships. The decision tree prioritized `compactness_mean_log` as the root splitting criterion. The initial split (< 0.2) separated the majority of benign case, subsequent splits utilized `perimeter_mean` and `texture_mean` to refine the classification.

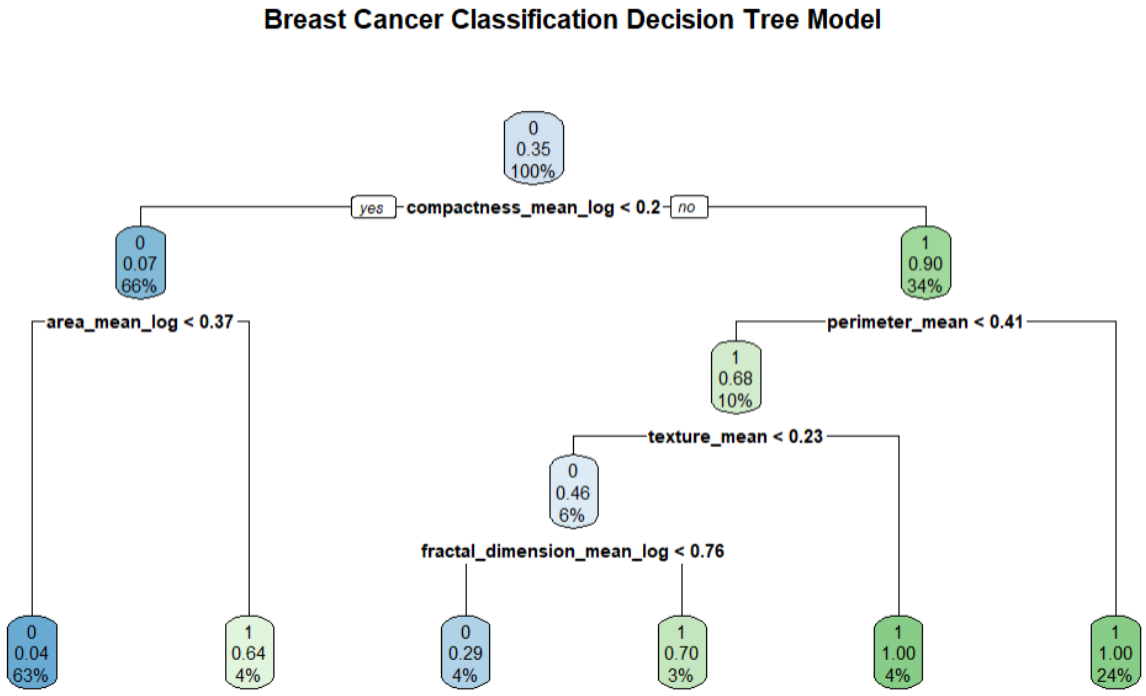


Figure 5: Decision Tree Model

Finally, a Random Forest ensemble of 500 decision trees. The model achieved an Out-of-Bag (OOB) error rate of 5.76%. Correctly identifying 250 benign and 126 malignant cases. However, the model had 13 malignant cases misclassified as benign compared to only 10 False Positives. The error rate for the malignant class is ~ 9.3 % and ~ 3.8% for benign (Table 3).

```
Call:
 randomForest(formula = as.factor(diagnosis) ~ ., data = train,      ntree = 500, mtry = floor(sqrt(ncol(train) - 1)), importance = TRUE)
               Type of random forest: classification
                     Number of trees: 500
No. of variables tried at each split: 3

        OOB estimate of  error rate: 5.76%
Confusion matrix:
    0   1 class.error
0 250  10  0.03846154
1  13 126  0.09352518
```

Table 3: Random Forest Model Result

Discussion and Conclusions

This study explored multiple classification approaches to predict breast cancer diagnosis using tumor measurement features. Multiple classification models were developed to predict breast cancer diagnosis using tumor measurement features. Logistic regression, decision tree, and random forest models were evaluated using accuracy, sensitivity, specificity, and AUC to assess both predictive performance and clinical relevance.

There were several, first, the dataset used in this analysis is relatively small and originates from a single source, which may limit the generalizability of the results to broader or more diverse patient populations. Second, the models were evaluated using a single train–test split; while sufficient for comparison, this approach may introduce variability depending on how the data are partitioned. Third, the study relied solely on existing tumor measurement features and did not incorporate additional clinical variables such as patient age, genetic markers, or medical history, which could improve predictive performance. Finally, threshold selection was fixed at 0.5, which may not be optimal in a medical context where minimizing false negatives is often more important than maximizing overall accuracy.

The decision tree model provided intuitive, rule-based classification and strong interpretability, making it useful for communication with clinicians. However, its predictive performance was slightly lower compared to logistic regression. The random forest model achieved competitive performance and demonstrated robustness to noise and complex interactions, but at the cost of reduced interpretability.

Overall, the results indicate that a well-specified logistic regression model remains a strong and reliable approach for breast cancer classification, balancing predictive accuracy, interpretability, and clinical applicability. Future work could incorporate cross-validation, threshold optimization to prioritize sensitivity, and regularized models to address multicollinearity. Expanding the dataset and validating results on external populations would further strengthen the clinical applicability of the findings.

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Appendix

The code for this project is available at:

https://github.com/Chung-Brandon/621-Data-Mining/blob/main/final.Rmd